

FORMALISATION AND EVALUATION OF PROPERTIES FOR CONSEQUENTIALIST MACHINE ETHICS

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INTRODUCTION

Motivation

- AI systems increasingly make decisions that impact humans' lives (self-driving cars)
- Need to follow ethical principles

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Problem

- Existing approaches lack rigorous justification and formal comparison.

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- consequentialism: judges actions only based on their **consequences**
- deontology: notion of **duty** to determine whether an action/ sequence of actions is right or wrong
- virtue ethics: uses agent's **character and intention** to determine the appropriateness of their actions

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We will focus on **consequentialism** because it is the dominant approach to machine ethics so far and the focus of this paper.

Question

How can machine ethics approaches properly validate their formalisation of ethical principles and determine the most appropriate principle(s) to apply in a given situation?

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Contributions

- Framework to evaluate ethical principles systematically,
- Define formal principles and properties for decision-making.

Running example

We have available to us exactly **one smart robot**, designed to assist in decision-making when “incidents” occur. A **fire just started** on the top floor of a building **full of employees**. The robot needs to **keep everyone safe** but also to **minimise the damage by extinguishing the fire**. It has three choices:

- (a) rush into the source of the fire and extinguish it,
- (b) personally help the evacuation of employees until everyone is safe,
- (c) trigger the fire alarm while calling the fire department.

What should it do?

SITUATION CALCULUS

Situation Calculus

What is it?

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Lets define the framework!

Definitions

Fluents

What they are: Properties of the world that can change over time,

Example: "The building is on fire", "Employees are safe", "The fire is extinguished",

Formal notation: $F(\vec{x}, s)$ means "property F with arguments $\vec{x}$ is true in situation s ."

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Situations

What they are: Snapshots of the world at a given time, including the history of actions that led to it,

Example:

- s_0 : Initial situation (building on fire, employees inside)
- $do(a, s)$: The situation after performing action a in situation s .

Formal notation: $do([evac, extg], s_0)$ means the situation after evacuating and then extinguishing the fire.

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Action Preconditions

What they are: Conditions that must be true for an action to be possible.

Example: To *extg*, the robot must be near the fire source.

Formal notation: $Poss(a(\vec{x}), s) \equiv \pi_a(\vec{x}, s)$ means "action a is possible in situation s if π_a is true."

Definitions

Effect Axioms

What they are: Rules that describe how actions change fluents.

- Positive effect: F becomes true after a .
- Negative effect: F becomes false after a .

Example:

- $do(evac, s)$ makes "employees are safe" true.
- $do(extg, s)$ makes "building is on fire" false.

Formal notation:

- $\gamma_F^+(\vec{x}, a, s) \rightarrow F(\vec{x}, do(a, s))$ (positive effect)
- $\gamma_F^-(\vec{x}, a, s) \rightarrow \neg F(\vec{x}, do(a, s))$ (negative effect)

Definitions

Successor State Axiom

What it is: A rule that fully describes how a fluent changes after an action,

Example: If *extg* is performed, "building is on fire" becomes false, and all other fluents stay the same unless changed,

Formal notation: $F(\vec{x}, do(a, s)) \equiv \gamma_F^+(\vec{x}, a, s) \vee (F(\vec{x}, s) \wedge \neg \gamma_F^-(\vec{x}, a, s))$

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Plans

What they are: Sequences of actions,

Example: $\pi = [evac, extg, call]$,

Formal notation: $do(\pi, s_0)$ is the situation after executing plan π from s_0 .

Updated Example

Consider an initial state of our example where the building is on fire and all employees are inside. The robot has options to extinguish the fire (extg), evacuate all employees (evac), and call the fire department (call). The employees will get injured if not evacuated immediately and the building will become more damaged the longer it is not extinguished.

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Consider an initial state of our example where the building is on fire and all employees are inside. The robot has options to extinguish the fire (extg), evacuate all employees (evac), and call the fire department (call). The employees will get injured if not evacuated immediately and the building will become more damaged the longer it is not extinguished.

Let's try and give some examples using the previous definition!

Example

In s_0 , we can say that:

- $onfire(s_0) \wedge inside(emp, s_0)$
- $Poss(call, s) \equiv \top$
- $Poss(extg, s) \equiv onfire(s)$
- $Poss(evac, s) \equiv inside(emp, s)$

Building is on fire, employees are inside, we can always call the fire department, we can only extinguish if there is a fire, and we can only evacuate if employees are inside.

FORMALISATION OF ETHICAL PRINCIPLES

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Overview

Objective here is not to achieve a **goal**, but rather to see if a plan satisfies an **ethical principle**.

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Goodness

We define a plan as **ethically permissable** based on its *goodness*.
We introduce the function *goodness(state(s))* which returns a *value*.
We also introduce *harmful(φ)* which is true if φ is a harmful fluent.

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We introduce the function $goodness(state(s))$ which returns a *value*.
We also introduce $harmful(\varphi)$ which is true if φ is a harmful fluent.

We simplify the notation for *admissible* plans and goodness:

- $state(\pi) = state(do(\pi, s_0))$
- $goodness(\pi) = goodness(state(\pi))$

Admissibility

Ethical Admissibility

We define a plan π as **ethically admissible** if it is both possible and satisfies an ethical principle.

The paper defines 5 ethical principles: **utilitarianism**, **hedonism**, **no new harm**, **remove harm** and **harm avoidance**.

Let's define them formally!

Ethical Principles

Utilitarianism

It focuses on bringing the greatest good to the greatest number of people:

$$\mathcal{EP}_{util}(\Pi, s_0) = \{\pi \in \Pi \mid goodness(\pi) \geq goodness(\pi') \text{ for all } \pi' \in \Pi\}$$

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Example:

If $\text{goodness}([\text{evac}, \text{extg}]) > \text{goodness}([\text{extg}, \text{call}])$, then $[\text{evac}, \text{extg}]$ is ethically permissible.

Note: $\mathcal{EP}$ stands for ethical permissibility function.

Ethical Principles

Hedonism

Based on the overall consequences that it produces, a direct interpretation of consequentialism: any plan is ethically permissible if it has more goodness at the end state compared to the initial state.

$$\mathcal{EP}_{hedon}(\Pi, s_0) = \{\pi \in \Pi \mid goodness(\pi) > goodness(\emptyset)\}$$

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Example:

If $goodness([evac]) \geq goodness(\emptyset)$, then $[evac]$ is ethically permissible.

Ethical Principles

No New Harm

It dictates that a plan should not cause any new harm, so an agent should not be directly(or actively) responsible for any harm. This also means that we tolerate a harm that is unavoidable.

$$\mathcal{EP}_{\text{nnHarm}}(\Pi, s_0) = \{\pi \in \Pi \mid \text{for all } \varphi, \text{ if } \text{harmful}(\varphi) \text{ and } \text{state}(s_0) \not\models \varphi, \\ \text{then } (\text{state}(\pi) \not\models \varphi \text{ or for all } \pi' \in \Pi, \text{state}(\pi') \models \varphi)\}$$

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$$\mathcal{EP}_{nnHarm}(\Pi, s_0) = \{\pi \in \Pi \mid \text{for all } \varphi, \text{ if } harmful(\varphi) \text{ and } state(s_0) \not\models \varphi, \\ \text{then } (state(\pi) \not\models \varphi \text{ or for all } \pi' \in \Pi, state(\pi') \models \varphi)\}$$

Explanation:

For every harmful predicate φ (like $harmful(injured(emp))$),

If φ is not true in the initial state ($state(s_0) \not\models \varphi$), Then φ must not become true after executing π ($state(\pi) \not\models \varphi$), unless φ becomes true after every admissible plan (for all $\pi' \in \Pi$, $state(\pi') \models \varphi$), i.e unavoidable.

Remove Harm

Remove Harm

A plan is ethically permissible if it removes existing harm in the initial situation. It requires the agent to mitigate harm whenever possible.

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Explanation:

For every harmful predicate φ , if it is found in the initial state ($\text{state}(s_0) \vdash \varphi$), then an acceptable plan π must remove it, or every possible plan must have it at the end (so unavoidable).

Harm Avoidance

Harm Avoidance

A plan is ethically permissible if it minimises the amount of harm in the end state.

$$\mathcal{EP}_{\text{harmAvoid}}(\Pi, s_0) = \{\pi \in \Pi \mid \text{for all } \varphi, \text{ if } \textit{harmful}(\varphi), \\ \text{then } (\textit{state}(\pi) \not\models \varphi \text{ or for all } \pi' \in \Pi, \textit{state}(\pi') \vdash \varphi)\}$$

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Explanation:

For every harmful predicate φ , an acceptable plan π must ensure that φ is not true in the end state, or every possible plan must have it at the end (so unavoidable). It is a combination of the two previous principles.

Example

Let's define that the most important thing is to keep the employees safe.
So we define the world as:

- $inside(emp, do(a, s)) \equiv inside(emp, s) \wedge a \neq evac$
- $onfire(do(a, s)) \equiv onfire(s) \wedge a \neq extg$
- $injured(emp, do(a, s)) \equiv inside(emp, s) \wedge onfire(s) \wedge a \neq evac$
- $injured'(emp, do(a, s)) \equiv injured(emp, s) \wedge a \neq evac \wedge inside(emp, s) \wedge onfire(s)$

Here emp is the employee and $injured(emp)$ the penalty of letting someone get injured. $injured'(emp)$ means even more damage to an employee.

Example

According to our previous definitions, we model the goodness function as:

- $goodness([call]) = -10$
- $goodness([extg]) = -10$
- $goodness([evac]) = 0$
- $goodness([call, evac]) = -10$
- $goodness([evac, call]) = 0$
- $goodness([extg, call]) = -20$
- $goodness([call, extg]) = -20$
- $goodness([evac, extg]) = 0$
- $goodness([extg, evac]) = -10$
- $goodness([call, evac, extg]) = -10$
- $goodness([evac, call, extg]) = 0$
- $goodness([extg, call, evac]) = -20$
- $goodness([call, extg, evac]) = -20$
- $goodness([evac, extg, call]) = 0$
- $goodness([extg, evac, call]) = -10$

Therefore, with the goodness of initial state set to 0, we have:

$$\begin{aligned}\mathcal{EP}_{util}(\Pi, s_0) &= \{[evac], [evac, call], [evac, call, extg], \\ &\quad [evac, extg], [evac, extg, call]\} \\ \mathcal{EP}_{hedon}(\Pi, s_0) &= \{\}\end{aligned}$$

Note: Hedonism returns empty set because no plan has a goodness strictly greater than the initial state, which has a goodness of 0.

Note: But there is a problem because lack of temporality (we will mention this later).

FORMALISATION OF PROPERTIES

Definition

Overview

We want to define properties that can be used to **evaluate and compare** ethical principles.

Based on which properties are verified, we can make an **educated choice** on what principle to follow according to the context/situation.

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The authors define 8 properties which are directly derived from the principles we just saw.

7 more are presented, and are derived from properties found in social choice theory (we will define what this means later).

Ethical Properties

Consequential Equivalence

We define this as moral actions that produce "good" consequences. It also states that two plans with the same consequences should be considered equally ethical.

If $\pi \in \mathcal{EP}(\Pi, s_0)$ and $state(do(\pi, s_0)) = state(do(\pi', s_0))$, then $\pi' \in \mathcal{EP}(\Pi, s_0)$, for all $\pi' \in \Pi$

Example:

Suppose $[evac, extg]$ and $[extg, evac]$ both result in the same final state (employees safe, fire extinguished). If $[evac, extg] \in \mathcal{EP}(\Pi, s_0)$, then $[extg, evac] \in \mathcal{EP}(\Pi, s_0)$.

Ethical Properties

Maximal Welfare

An ethically permissible plan must produce at least as much goodness as any other plan that is ethically admissible.

if $\pi \in \mathcal{EP}(\Pi, s_0)$, then for all $\pi' \in \Pi$, $goodness(\pi) \geq goodness(\pi')$

Example:

If $[evac, extg] \in \mathcal{EP}(\Pi, s_0)$ and $[call]$ is a plan (not yet determined if ethical or not), then for $[call]$ to be ethically permissible, it must have at least as much goodness as $[evac, extg]$.

Ethical Properties

Aggregationism

If a plan is ethically permissible and another plan is even better (i.e greater goodness), then the better plan must also be ethically permissible.

if $\pi \in \mathcal{EP}(\Pi, s_0)$ and $goodness(\pi') \geq goodness(\pi)$,
then $\pi' \in \mathcal{EP}(\Pi, s_0)$ and for all $\pi' \in \Pi$,

Ethical Properties

Independance

The permissibility of a plan should not depend on irrelevant alternatives. If a plan is permissible in a larger set, it should remain permissible in any subset containing it.

$$\mathcal{EP}(\Pi, s_0) \cap \Pi' \subseteq \mathcal{EP}(\Pi', s_0)$$

Intuition:

If a plan Π is ethically permissible, and we consider the intersection between this plan and another set of plans Π' , then this intersection should be a subset of the ethically permissible plans in Π' .

Beneficence

A permissible plan must be better than doing nothing. In other words, the overall goodness at the end of the plan must be better than the overall goodness at the beginning of the plan.

If $\pi \in \mathcal{EP}(\Pi, s_0)$, then $goodness(\pi) > goodness(\emptyset)$.

Note: this property and Independence are closely tied to Hedonism.

Ethical Properties

New Notations

We define the following terms for the next three properties:

- $harms(\pi) = \{\varphi \mid state(\pi) \vdash \varphi \wedge harmful(\varphi)\}$
- $newState(\pi) = \{F(\vec{x}) \mid F(\vec{x}, do(\pi, s_0)) \wedge \neg F(\vec{x}, s_0)\}$
- $oldState(\pi) = \{F(\vec{x}) \mid F(\vec{x}, do(\pi, s_0)) \wedge F(\vec{x}, s_0)\}$

Positive Responsibility

This property defines that ethical permissibility should be determined only by the new state such actions produce. Together with Harmlessness, this property is closely aligned with the No New Harm principle.

If $\pi \in EP(\Pi, s_0)$ and $newState(\pi) = newState(\pi')$,
then $\pi' \in EP(\Pi, s_0)$, for all $\pi' \in \Pi$.

Ethical Properties

Negative Responsibility

The property is requiring that an ethically plan should eliminate existing harm or maintain the current good, and is closely aligned with the Remove Harm principle. I.e we want the agent to be responsible for inaction or maintaining harm.

If $\pi \in \mathcal{EP}(\Pi, s_0)$ and $oldState(\pi) = oldState(\pi')$,
then $\pi' \in \mathcal{EP}(\Pi, s_0)$, for all $\pi' \in \Pi$.

Ethical Properties

Harmlessness

This property focuses on the harm of a plan. We should only allow plans that aim to prevent harm, and so the harm at the end of the plan determines its ethical permissibility.

If $\pi \in \mathcal{EP}(\Pi, s_0)$ and $harms(\pi') \subseteq harms(\pi)$,
then $\pi' \in \mathcal{EP}(\Pi, s_0)$, for all $\pi' \in \Pi$.

Social Choice Properties

Overview

Social choice theory designs protocols to select the best outcomes from preferences or choices.

- Aggregating individual preferences into collective decisions,
- Analyzing voting systems,
- Investigating choice functions.

The paper presents seven social choice properties to ensure $\mathcal{EP}$ is robust and consistent.

Consistency Preservation (cp)

This property ensures that if there are any admissible plans, at least one must be ethically permissible. It guarantees that a decision is always made, even in dilemmas.

If $\Pi \neq \emptyset$, then $\mathcal{EP}(\Pi, s_0) \neq \emptyset$.

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Iteration (it)

This property requires that applying the ethical permissibility function twice (or more) does not change the result. It ensures stability in decisions.

$$\mathcal{EP}(\mathcal{EP}(\Pi, s_0), s_0) = \mathcal{EP}(\Pi, s_0).$$

Cut (c)

This property states that if you remove some impermissible plans from Π , the permissibility of the remaining plans does not change. It ensures robustness when the set of plans is reduced.

If $\mathcal{EP}(\Pi, s_0) \subseteq \Pi' \subseteq \Pi$, then $\mathcal{EP}(\Pi, s_0) \subseteq \mathcal{EP}(\Pi', s_0)$.

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Aizerman (aiz)

This property ensures that if you restrict the set of plans to a subset Π' that includes all permissible plans, the result does not change. It guarantees that adding back previously rejected plans does not affect the permissibility of the kept plans.

If $\mathcal{EP}(\Pi, s_0) \subseteq \Pi' \subseteq \Pi$, then $\mathcal{EP}(\Pi', s_0) \subseteq \mathcal{EP}(\Pi, s_0)$.

Note: these two are tightly related, and refer to contextual blindness.

Chernoff (ch)

This property requires that if you shrink the set of plans ($\Pi' \subseteq \Pi$), the permissible plans in Π' must include those that were permissible in Π . It ensures fairness when the set of plans is reduced.

If $\Pi' \subseteq \Pi$, then $\mathcal{EP}(\Pi, s_0) \cap \Pi' \subseteq \mathcal{EP}(\Pi', s_0)$.

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If $\Pi' \subseteq \Pi$, then $\mathcal{EP}(\Pi, s_0) \cap \Pi' \subseteq \mathcal{EP}(\Pi', s_0)$.

Sen (s)

This property states that if a plan π is permissible in both Π and Π' , then it must be permissible in their intersection $\Pi \cap \Pi'$. It ensures consistency across different contexts.

If $\mathcal{EP}(\Pi, s_0) \cap \mathcal{EP}(\Pi', s_0) \neq \emptyset$, then
 $\mathcal{EP}(\Pi \cap \Pi', s_0) \subseteq \mathcal{EP}(\Pi, s_0) \cap \mathcal{EP}(\Pi', s_0)$.

Arrow (iia)

This property, also known as Independence of Irrelevant Alternatives, requires that if a subset Π' of plans includes some permissible plans from Π , then the permissible plans in Π' are exactly those. It ensures that the permissibility of a plan depends only on how it compares to other available plans.

If $\Pi' \subseteq \Pi$ and $\mathcal{EP}(\Pi, s_0) \cap \Pi' \neq \emptyset$, then $\mathcal{EP}(\Pi, s_0) \cap \Pi' = \mathcal{EP}(\Pi', s_0)$.

RESULTS

Results

Recap

Principles are **filters**, which directly select which plans are permissible based on their ethical criteria.

Properties are tools to **evaluate and compare principles**: how a principle behaves or what it guarantees.

Example:

- Maximal Welfare: "Does this principle always select plans that maximize goodness?"
- Harmlessness: "Does this principle avoid causing unnecessary harm?"
- Consistency Preservation: "Does this principle always select at least one plan, even in dilemmas?"

Results

	Utilitarianism	Hedonism	No New Harm	Remove Harm	Harm Avoidance
Consequential Equivalence	✓	✓	✓	✓	✓
Maximal Welfare	✓	✗	✗	✗	✗
Aggregationism	✓	✓	✗	✗	✗
Independence	✗	✓	✗	✗	✗
Beneficence	✗	✓	✗	✗	✗
Positive Responsibility	✗	✗	✓	✗	✗
Negative Responsibility	✗	✗	✗	✓	✗
Harmlessness	✗	✗	✓	✓	✓
Consistency Preservation	✓	✗	✗	✗	✗
Iteration	✓	✓	✓	✓	✓
Cut	✓	✓	✓	✓	✓
Aizerman	✓	✓	✗	✗	✗
Chernoff	✓	✓	✓	✓	✓
Sen	✓	✓	✓	✓	✓
Arrow	✓	✓	✓	✓	✓

Table 1: Properties of different ethical principles: ✓ indicates satisfaction and ✗ indicates non-satisfaction

Utilitarianism

Key Idea

Utilitarianism focuses on **maximizing overall welfare**, even if it means accepting some harm or less-than-ideal outcomes.

"The greatest good for the greatest number."

Consequential Equivalence

Plans with the same outcome are treated equally.

[*evac*, *extg*] and [*extg*, *evac*] are both permissible if they result in the same state (employees safe, fire extinguished).

Maximal Welfare

A permissible plan must produce at least as much goodness as any other plan.

If [*evac*, *extg*] has *goodness* = 10 and [*call*] has *goodness* = 5,

only [*evac*, *extg*] is permissible.

Aggregationism

If a plan is permissible and another plan is even better, the better plan must also be permissible.

If [*evac*] is permissible (*goodness* = 7),

[*evac*, *extg*] (*goodness* = 10) must also be permissible.

Hedonism

Key Idea and Contrast

Hedonism focuses on **ensuring a net benefit** compared to doing nothing.

"A plan must be better than inaction."

Consequential Equivalence & Aggregationism

Same as Utilitarianism.

Independence

Permissibility does not depend on irrelevant alternatives.

If $[evac]$ is permissible in $\Pi = \{[evac], [extg], [call]\}$,
it remains permissible in $\Pi' = \{[evac], [extg]\}$.

Beneficence

Permissible plans must improve the initial state.

$[call]$ ($goodness = -2$) is not permissible if the initial state has $goodness = 0$.

No New Harm

Key Idea

No New Harm focuses on **avoiding new harm**, even if it means tolerating existing harm.

"Do not produce a harm if it can be avoided."

Consequential Equivalence

Plans with the same outcome are treated equally.

[*evac*, *call*] and [*call*, *evac*] are both permissible if they result in the same state (no new harm introduced).

Positive Responsibility

A permissible plan must not introduce new harm.

If [*evac*] avoids new harm but [*extg*] risks injuries, only [*evac*] is permissible.

Harmlessness

If a plan causes less harm than a permissible plan, it must also be permissible.

If [*evac*] is permissible, [*evac*, *call*] (less risky) must also be permissible.

Remove Harm

Key Idea

Remove Harm focuses on **eliminating existing harm**, even if it means tolerating some new harm.

"Eliminate a pre-existing harm if possible."

Consequential Equivalence

Plans with the same outcome are treated equally.

[*extg*, *call*] and [*call*, *extg*] are both permissible if they result in the same state (existing harm removed).

Negative Responsibility

A permissible plan must eliminate existing harm.

If [*extg*] removes the fire but [*call*] doesn't,
only [*extg*] is permissible.

Harmlessness

If a plan causes less harm than a permissible plan,
it must also be permissible.

If [*extg*] is permissible, [*extg*, *evac*] (removes more harm) must also be permissible.

Harm Avoidance

Key Idea and Contrast

Harm Avoidance focuses on **avoiding all avoidable harms**, both new and existing.

"Avoid all avoidable harms!"

- Stricter than No New Harm and Remove Harm.
- May reject all plans if none avoid all harms.

Consequential Equivalence

Plans with the same outcome are treated equally.

[*evac*, *extg*] and [*extg*, *evac*] are both permissible if they result in the same state (all harms avoided).

Harmlessness

A permissible plan must avoid all avoidable harms.

If [*evac*] leaves fire burning and [*extg*] risks injuries, neither may be permissible.

Social Choice Properties: Universally Satisfied

Key Idea

These properties are satisfied by **all five principles**:

Iteration

Applying the ethical permissibility function twice doesn't change the result.

Example: If $[evac, extg]$ is permissible, re-applying $\mathcal{EP}$ won't exclude it.

Cut

Removing impermissible plans doesn't affect permissible ones.

Example: If $[evac]$ is permissible in Π , it remains permissible in $\Pi' = \{[evac], [extg]\}$.

Chernoff

Shrinking the set of plans preserves permissibility.

Example: If $[evac]$ is permissible in Π , it remains permissible in $\Pi' = \{[evac]\}$.

Sen

Plans permissible in both sets must be permissible in their intersection.

Example: If $[evac]$ is permissible in both Π and Π' , it's permissible in $\Pi \cap \Pi'$.

Arrow

Permissibility depends only on available plans.

Example: If $[evac]$ is permissible in $\Pi = \{[evac], [extg]\}$, its permissibility doesn't change if $[call]$ is added.

Consistency Preservation: Utilitarianism Only

Key Idea

Only Utilitarianism satisfies this property:

Definition

If there are any admissible plans, at least one must be permissible.

If $\Pi \neq \emptyset$, then $\mathcal{EP}(\Pi, s_0) \neq \emptyset$.

Why Utilitarianism?

Always selects the "least bad" option, even if all plans are harmful.

Example: In $\Pi = \{[evac], [extg]\}$ where both cause harm, Utilitarianism picks the one with less harm.

Why Not Others?

Harm-based principles may reject all plans if all cause harm.

Example: No New Harm might reject both $[evac]$ and $[extg]$ if both introduce new harm.

Aizerman: Only Hedonism and Utilitarianism

Key Idea

Only these Utilitarianism and Hedonism satisfy Aizerman.

Definition

Restricting plans to a subset containing all permissible plans doesn't change results.
If $\mathcal{EP}(\Pi, s_0) \subseteq \Pi' \subseteq \Pi$, then $\mathcal{EP}(\Pi', s_0) \subseteq \mathcal{EP}(\Pi, s_0)$.

Why Utilitarianism?

Focuses on maximal welfare, unaffected by adding back rejected plans.

Example: If $[evac]$ is permissible in Π , it remains so in $\Pi' = \{[evac], [do\ nothing]\}$.

Why Hedonism?

Only requires improvement over inaction, unaffected by additional options.

Example: If $[evac]$ improves over inaction, it remains permissible even if $[do\ nothing]$ is added.

Why Not Others?

Harm-based principles may change permissibility when new options are added.

Example: Harm Avoidance might reject $[evac]$ if $[do\ nothing]$ is added (avoids all harm).

Conclusion: Novel Contributions

Key Innovations

- **First formal framework** for consequentialist machine ethics using situation calculus.
- **Two major contributions:**
 - Formalized 5 ethical principles
 - Defined 15 properties
- **Proved property satisfaction** for each principle, enabling precise characterization.
- **Bridged philosophy and AI** by translating ethical theories into computational logic.

Proofs are available here: bit.ly/3JUVrU9

Why This Matters

- Enables **rigorous analysis** of ethical AI systems.
- Provides **toolkit** to validate and compare ethical approaches.
- Facilitates **transparent, explainable** ethical decision-making in AI.

Future Directions and Impact

Conclusions

- No single principle is perfect - each satisfies some properties but not others.
- **Trade-offs** must be considered based on application context.

Future Research Opportunities

- Extend to temporal logic (big problem in my opinion).
- Study **prioritization of principles** in complex scenarios (with time!).
- Validate with **real-world case studies**.

THANK YOU

APPENDIX

Full Example Definition

Fluents

- *OnFire(s)*
- *EmployeesSafe(s)*
- *FireExtinguished(s)*
- *FireDepartmentCalled(s)*
- *BuildingDamaged(s)*
- *EmployeesInjured(s)*

Actions

- *extg*
- *evac*
- *call*

Full Example Definition

Initial Definition

- $OnFire(s_0) = True$
- $EmployeesSafe(s_0) = False$
- $FireExtinguished(s_0) = False$
- $FireDepartmentCalled(s_0) = False$
- $BuildingDamaged(s_0) = False$
- $EmployeesInjured(s_0) = False$

Action Precondition Axioms

- $Poss(extg, s) \equiv OnFire(s)$
- $Poss(evac, s) \equiv \neg EmployeesSafe(s)$
- $Poss(call, s) \equiv \neg FireDepartmentCalled(s)$

Positive Effect Axioms

- $\gamma_{FireExtinguished}^+(extg, s) \rightarrow FireExtinguished(do(extg, s))$
- $\gamma_{EmployeesSafe}^+(evac, s) \rightarrow EmployeesSafe(do(evac, s))$
- $\gamma_{FireDepartmentCalled}^+(call, s) \rightarrow FireDepartmentCalled(do(call, s))$

Negative Effect Axioms

- $\gamma_{OnFire}^-(extg, s) \rightarrow \neg OnFire(do(extg, s))$
- $\gamma_{BuildingDamaged}^-(call, s) \rightarrow \neg BuildingDamaged(do(call, s))$
- $\gamma_{EmployeesInjured}^-(evac, s) \rightarrow \neg EmployeesInjured(do(evac, s))$